Assignment 1

Question 1

Functional and Non-Functional Requirements in Databases:

Objective: Analyze a system to determine the necessary database requirements. For each system listed below, identify both the functional and non-functional requirements of the database.

Systems:

- Library System.
- Bank System.
- University System.
- Restaurant System.
- Supermarket System.
- ATM System.
- Job Boards System (Companies that offer job vacancies).
- Hospital System.
- Car Rental System.
- Hotel System.
- Nursery System.
- Clinics System.
- Food Delivery System.
- Real Estate System.
- Pharmacy System.

Instructions:

1. Select one system from the list.
2. Identify **at least 20 functional requirements** that the database should fulfill.
3. Identify **at least 5 non-functional requirements** for the database of the chosen system.
4. Provide a brief justification for each requirement based on the system's needs.

PLEASE NOTE THAT:

- Requirements must be categorized using the **MOSCOW** technique and each requirement should have a unique **ID**.

- The Requirements Specs should be submitted in a **word** or **pdf file** and **uploaded to git/ git extension repository** that you create and will be used through the course.

- The Assignment Submission on Moodle should contain both **The file** and **the link to the Git repository**, you need to make sure that it is accessible by non-owners.

Question 2

- Using the functional requirements identified above for your system, **design the database structure by creating an Entity-Relationship Diagram (ERD).**

Instructions:

1. Select the system you worked on in the previous question.
2. Based on the functional requirements you identified, create an Entity-Relationship Diagram (ERD) to visualize the database structure for the system.
3. The ERD should clearly represent:
    - Entities: The main objects in your system (e.g., Users, Orders, Products, etc.).
    - Attributes: Key data points for each entity (e.g., Name, Date, ID, etc.).
    - Relationships: How the entities relate to each other (e.g., one-to-one, one-to-many, many-to-many).
4. **The ERD SHOULD BE HANDWRITTEN, scanned PDF**, and uploaded to Moodle.